Renan Souza, Ph.D.

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Summary

Research staff scientist working on workflow provenance, data integration, and agentic AI workflows for science. My work focuses on making large-scale, AI-driven workflows transparent, reliable, and reproducible, including when AI agents influence downstream outcomes. I build real systems and validate them in active scientific campaigns.

Areas of Expertise

Workflow provenance and reproducibility in large-scale science ◆ Provenance-aware data management for the AI lifecycle ◆ Agentic AI workflows with traceability and reliability ◆ Edge-Cloud-HPC workflow integration and federated operations ◆ Scalable workflow analytics and human-in-the-loop steering

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Education

Federal University of Rio de Janeiro, Brazil - Rio de Janeiro, Brazil

Ph.D. in Computer Science | Sep 2015 — Dec 2019 | COPPE/Data and Knowledge Engineering
Thesis: Supporting User Steering in Large-scale Workflows with Provenance Data
M.Sc. in Computer Science | Jan 2013 — Jul 2015 | COPPE/Data and Knowledge Engineering
B.Sc. in Computer Science | Jan 2009 — Dec 2012

International experience:

Visiting Ph.D. Student - Inria/Univ. Montpellier, France | Jan 2019 — Mar 2019 (Ph.D.)
Computer Science exchange student - Missouri State University, U.S. | Jun 2011 — Jun 2012 (B.Sc.)

Experience

Oak Ridge National Laboratory — **Oct 2022 — Present**
Staff Scientist & Sr. Software Engineer, HPC Workflows, Data & AI — **Knoxville, USA**

- Led R&D on workflow provenance and observability for AI-driven science, focusing on transparency, reliability, and reproducibility in end-to-end workflows.
- Developed provenance models and system mechanisms to connect user intent, agent decisions, workflow executions, and downstream results in unified traces.
- Validated methods through real deployments spanning Edge-Cloud-HPC environments and interactive workflows that require low-latency, auditable responses.
- Published and presented results in workflow and eScience venues, and drove community engagement through tutorials and reference architectures.

IBM Research — **Apr 2015 — Oct 2022**
Staff Scientist & Sr. Software Engineer, Cloud, Data & AI — **Rio de Janeiro, Brazil**

- Conducted applied R&D in data management and AI systems, producing peer-reviewed outputs and patented innovations.
- Explored scalable data services and governance-aware approaches that support AI lifecycle requirements in enterprise contexts.
- Collaborated internationally across research and engineering teams to validate ideas in real deployments and user scenarios.

SLAC National Accelerator Laboratory, Stanford University — **May 2013 — Dec 2014**
Research Software Engineering Intern — **Menlo Park, USA**

- Applied semantic web and scalable data management methods to publish structured measurement data for broad community use.

Federal University of Rio de Janeiro — **Jan 2010 — Sep 2014**
Software Engineer (Intern → Engineer) — **Rio de Janeiro, Brazil**

- Built applied semantic web and linked data solutions, grounding research ideas in real systems and user needs.
- Early applied work on integrating heterogeneous data sources for analytics and reporting.

Petrobras — **May 2007 — May 2008**
IT Intern — **Rio de Janeiro, Brazil**

- Early industry experience in software delivery and operations.

Selected on-going Projects

American Science Cloud (AmSC)

A secure, federated cloud environment integrating DOE facilities and data to enable AI-ready datasets and scalable model services. I develop multi-agent methods and implementations within ORNL use cases for cross-facility science workflows.

Orchestrated Platform for Autonomous Laboratories (OPAL)

A DOE multi-lab initiative to make biological discovery self-driving using AI, robotics, and automated experimentation. I lead agentic AI systems that connect interactive intent to Frontier-scale execution and multimodal analysis.

Advanced Manufacturing into Leadership-class Supercomputers via AI Agents

A modular architecture for autonomous cross-facility experimentation using AI agents, programmable facility APIs, and provenance-aware workflows. I led the multi-agent communication and workflow steering components.

Flowcept

A provenance system for traceable, auditable, and reproducible agentic workflows that unifies runtime signals, lineage, and agent interactions. I created and lead Flowcept, and we are advancing agentic provenance capabilities.

Technical Knowledge

Programming Languages: Python, Java, C, C++, C#, Shell, NodeJS, Scala, Lua
Data Science/ML: PyTorch, MLFlow, Airflow, Pandas, Polars, Jupyter, Matplotlib, Plotly
Agentic AI: MCP, CrewAI, LangChain, Streamlit, Chainlit, RAG, LLM-based orchestration
Big Data & Streaming Platforms: Apache Spark, Dask, Kafka, Redis, RabbitMQ
Parallel & Distributed Programming: MPI, OpenMP, CUDA, PubSub
Cloud, HPC, DevOps: Kubernetes, OpenShift, Docker, Slurm, LSF, PBS; CI/CD GitHub Actions, Jenkins, Travis; Grafana
Databases & Knowledge Graphs: PostgreSQL/PostGIS, MySQL, MongoDB, Elasticsearch, HBase, Hive, Redis, LMDB, Polystores; AllegroGraph, Jena, Virtuoso, RDF, SPARQL, OWL

Selected Publications

For 50+ published papers and 10+ patents, visit https://renansouza.org/publications

- **R. Souza**, T. J. Skluzacek, S. R. Wilkinson, M. Ziatdinov, and R. F. da Silva, "Towards Lightweight Data Integration using Multi-workflow Provenance and Data Observability," in IEEE International Conference on e-Science, 2023, doi: 10.1109/e-Science58273.2023.10254822.
- **R. Souza**, L. G. Azevedo, V. Lourenço, E. Soares, R. Thiago, et al., "Workflow Provenance in the Lifecycle of Scientific Machine Learning," Concurrency and Computation: Practice and Experience, 2021, doi: 10.1002/cpe.6544.
- **R. Souza**, A. Gueroudji, S. DeWitt, D. Rosendo, T. Ghosal, et al., "PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows," in IEEE International Conference on e-Science, 2025.
- **R. Souza**, S. Caino-Lores, M. Coletti, T. J. Skluzacek, A. Costan, et al., "Workflow Provenance in the Computing Continuum for Responsible, Trustworthy, and Energy-Efficient AI," in IEEE International Conference on e-Science, 2024, doi: https://doi.org/10.1109/e-Science62913.2024.10678731.